HYDROWORX

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350 Series User Guide and Owner Manual



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By accepting this Manual, the Recipient agrees to keep confidential the information contained herein or made available by the Company in connection with any service, maintenance, or repair of the HydroWorx System. By accepting this Manual, the Recipient also agrees to make no copies of the Manual in whole, in part, or in any form and to use this Manual solely for the purposes of understanding the proper use of the HydroWorx System.

Limited Warranty

HydroWorx International, Inc. (HydroWorx) warrants solely to the original Buyer that this product, consisting of the treadmill system, jet system, electronic controls, and water circulation equipment will be free from defects in materials and workmanship, when given normal, proper and intended installation and usage, for the period set forth below from the date noted on signed start-up completion form.

LIMITED WARRANTY PERIOD	COMPONENTS COVERED
One (1) Year	▪ Water Circulation Equipment (heater, filter housing, brominator and pump)* ▪ Badu Jet and Pump* ▪ Treadmill Motor ▪ Treadmill Seal and Bearings* ▪ Treadmill System* ▪ Treadmill Belt* ▪ Control System ▪ HMI System ▪ Water Storage Tank
Five (5) Year	▪ Treadmill Chamber Structural Integrity

*Proper maintenance and lubrication schedules must be followed to keep the warranty intact.

During the limited warranty period, HydroWorx will repair or replace, at its option and without cost to the Buyer, any portion of the product which is confirmed by HydroWorx to be defective.

This warranty shall not apply to any expendable component, such as, but not limited to chemicals, fuses, filters, and lubricants. HydroWorx shall have no obligations to make repairs or replacements which result, in whole or in part, from (I) catastrophe, fault or negligence of Buyer, (II) improper or unauthorized use of product, (III) maintenance or installation of the product which is not in full

compliance with the requirements of the HydroWorx Operating Manual and Software Manual, (IV) the presence of debris or silt in the product, (V) the failure to maintain water cleanliness and chemistry, accordance with the recommendations of circulation equipment suppliers, (VI) any modification of the product (including, but not limited to, removal or affixing any object to the equipment or removing any component part of the product without prior specific written instruction from HydroWorx), or (VII) causes external to the product such as, but not limited to, power failure.

In the event after inspection by HydroWorx the product proves not to be defective or is not subject to the warranty protection because of normal use, wear and tear, misuse, lapse of time, accident, modification, or any other exclusion specified above, the Buyer will be charged for repair or replacement services at HydroWorx's standard rates then in effect together with the "portal to portal" expenses incurred by HydroWorx in providing the service.

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Part

1

User Guide

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1.1 Safety While Using the HydroWorx 350

- Never operate the door latch yourself, unless you received training from the person supervising you, or unless you have read the full operating instructions.

- It is advisable to wear the safety tether around your wrist.

- You should start at the lowest speed before increasing the speed.

- You can stand astride the treadmill on the side platforms. Once the treadmill is running, you can gradually step onto the belt while letting the handrails take most of your weight. You can use the same technique to step off the treadmill onto the side platforms.

- Remember that the water takes approximately 3 minutes to drain from maximum fill depth for the chamber. So if you begin to feel unwell in any way while exercising it is important that you stop immediately.

- You can stop the treadmill at any time by pressing the emergency stop button located below the HMI screen on the operator panel. This will decelerate and stop the treadmill, stop the water flowing from the resistance jet and drain the water from the chamber. Pulling the tether out of the switch does the same thing.

- Never attempt to climb out over the sides of the chamber.

- Never push or pull any of the glass panels.

Part 2

Operating Instructions

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2.1 Introduction

The HydroWorx System provides an ideal environment for reduced weight-bearing exercise, a personal hydro-therapy environment where the client's progress can be easily monitored.

2.1.1 350 Treadmill

The tempered glass panels on the stainless steel chamber allow a clear view of the user throughout the rehabilitation or training program.

Standing areas are provided on either side of the treadmill, enabling the user to step quickly and easily off the treadmill. The door incorporates a secure latching system, ensuring the safety of the user.



2.1.2 Control System (HMI)

The touch screen controls provide intuitive control of water depth, treadmill speed, and the resistance jet. Water depth and treadmill speed can be quickly adjusted to pre-set levels for convenience. Intermediate water levels and speeds can also be manually adjusted.



2.1.3 Plant Pack System

After initially filling the storage tank, water is constantly re-circulated and filtered both during use and when the system is idle. The system therefore has minimal water usage. Water quality is monitored and adjusted manually, water chemical treatment is achieved with an erosion feeder, and water is heated via electric heater.



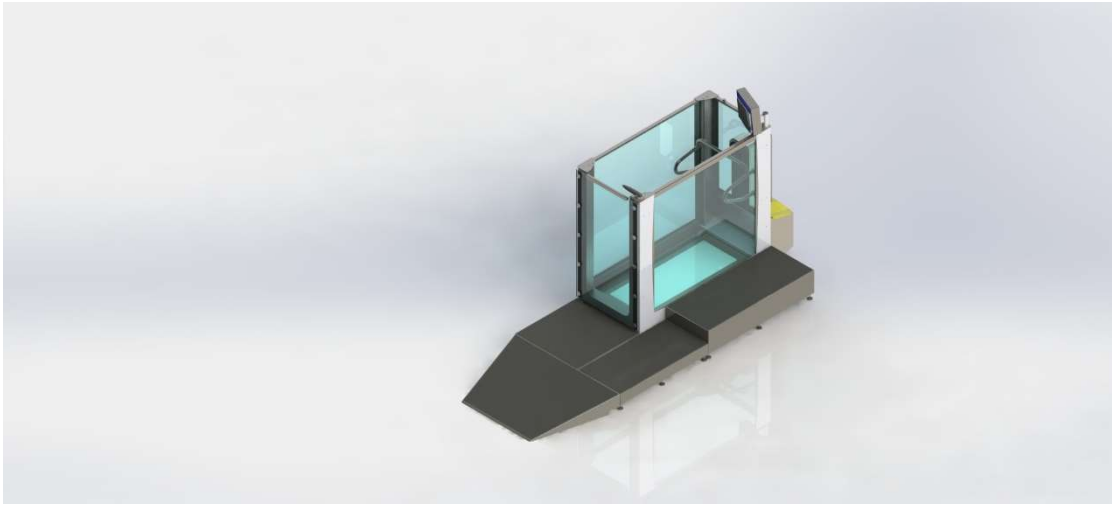
2.1.4 Water Storage Tank

The Water Storage Tanks hold up to 1,000 gallons of water and are used to transfer treated water to the 350 Treadmill via the Plant Pack.



2.1.5 Decking and Access Ramp

The optional decking is covered in a black non-slip water resistant floor covering with attractive hard-wearing stainless steel side trim. The access ramp is stainless steel with a non-slip covering.



2.2 Responsibilities of Users

Set out below is a guide to the responsibilities of each group of users.

2.2.1 Clients Undertaking Exercise or Treatment

- The client or patient using the 350 Treadmill MUST comply with all safety notices and the user guide.
- The user MUST NOT attempt to correct a fault or troubleshoot the HydroWorx System.

2.2.2 Maintenance Personnel

- Maintenance personnel must have read and understood these operating instructions and the Maintenance Instructions before using the HydroWorx System.
- Maintenance personnel are responsible for implementing the routine inspection and maintenance schedule laid out in the maintenance instructions.

2.3 Safety When Using the 350 Treadmill

2.3.1 Safety Tether

The 350 Treadmill is equipped with a safety tether which is recommended to be worn by all users. The tether is a wrist strap that is worn on the left wrist, with a cord that attaches to a cut-out switch mounted on the left hand side of the 350 Treadmill. If the tether is pulled with a force of more than 2.2LBS, it will activate the switch, which instructs the control system to decelerate, stop the treadmill, shut off the jet, and drain the 350 Treadmill.

The 350 Treadmill will drain from 56” to empty in 3 minutes. If the system is used for treating a medical condition then the person administering the treatment must take this drain time into consideration when assessing the risks to the individual using the chamber.

2.3.2 Emergency Stop (E-Stop)

Press the E-Stop located below the HMI screen on the control panel (as shown in figure 2-1) to stop the treadmill, shut off the jet, and drain the Trainer Tank. When pressed, the E-Stop push-button will latch in place. To reset the button, twist it in the direction indicated by the arrows on the button. This will allow it to pop out again.

Figure 2-1 E-Stop position



2.3 3 Protection of the Glass Panels



WARNING

In order to protect the glass panels form damage, users must not:

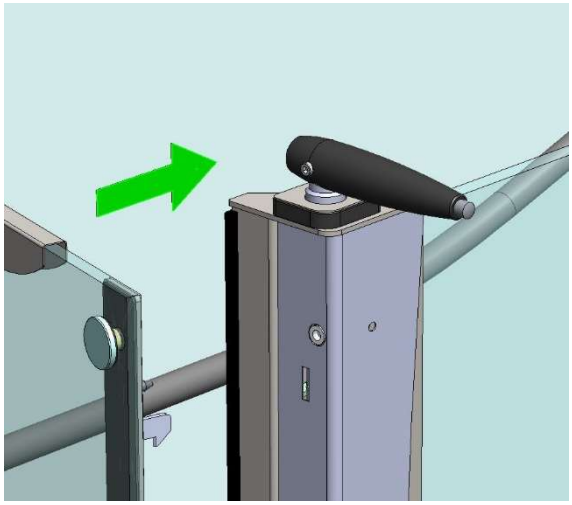
- Climb over the sides of the chamber or push the glass panels outwards
- Bring hard, sharp or metallic objects in or around the chamber
- Wear jewelry or watches when using the chamber

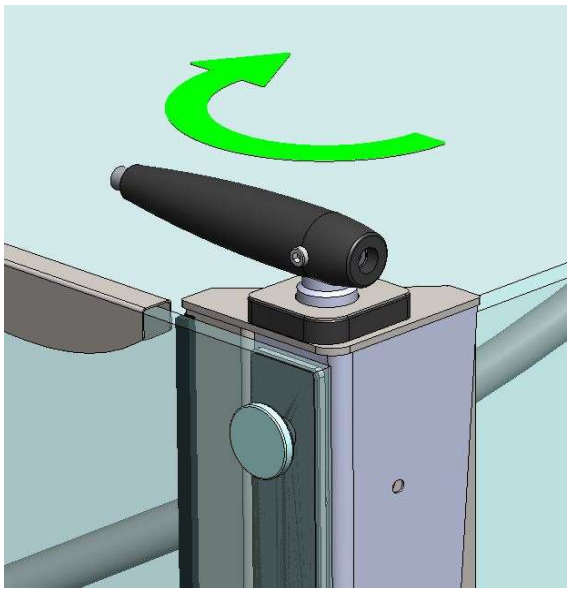
2.4 DOOR LATCH OPERATION

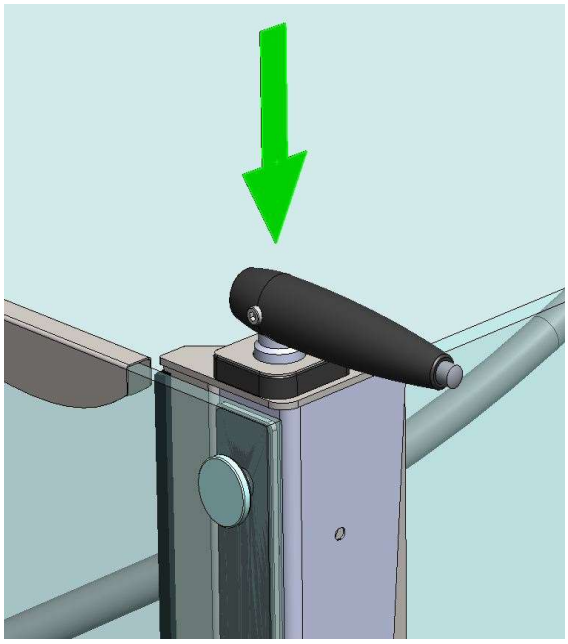
The door is latched shut by the handle mounted on the doorframe. Rotate the handle to clamp the door shut. Press it down to release the door. This section shows the door operating procedure.

Evacuate all water from the chamber before opening the door.

Graphic on HMI before water can enter 350 Treadmill


1. Closing The Door	
	**To start, push the door closed. The door will latch itself automatically, although it will only be loosely held closed at this point.**

Latching the door (Continued)	
	**Rotate the latch handle 180 degrees and assure the button on the end of the handle clicks out.** **NOTE:** **The door is not securely clamped against the door seal until the button on the end of the door handle clicks out.**

3. Opening the door	
	**To unlatch the door, first push in the button and rotate the handle 180 degrees counter-clockwise to the starting position. The door will un-clamp from the seal, but remain latched.** **To release the door latch, push the latch handle firmly downwards. The door will unlatch and spring open slightly.** **Pushing the handle down to release the door can only be done when the handle is in this position.**

NOTE: **If the door handle is rotated while the chamber is full of water, the safety latch will hold the door closed, water will however leak freely around the door seal. If this happens, quickly turn the handle clockwise to reseal the door. On no account press the handle down to release the door latch, as the door will open rapidly, allowing the water in the chamber to escape.**

Always leave the door closed when the system is not in use!

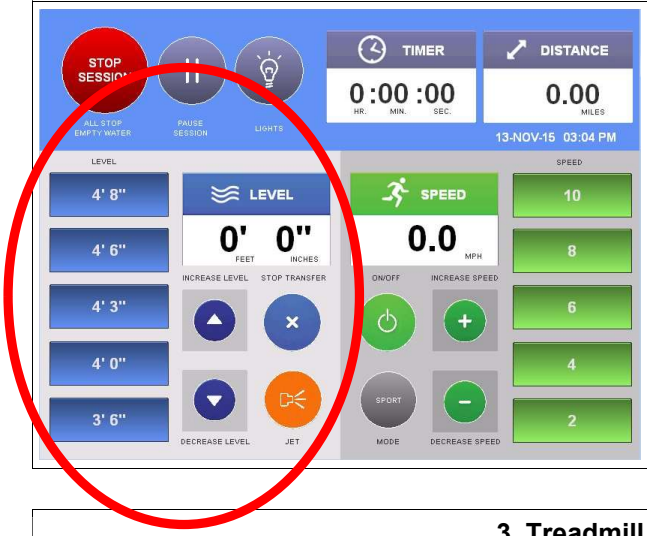
2.5 TOUCH SCREEN OPERATION

2.5.1 Procedures

The HydroWorx System is controlled using the touch-screen located on the 350 Treadmill. This section shows a typical sequence of operations for using the HydroWorx System touch-screen.

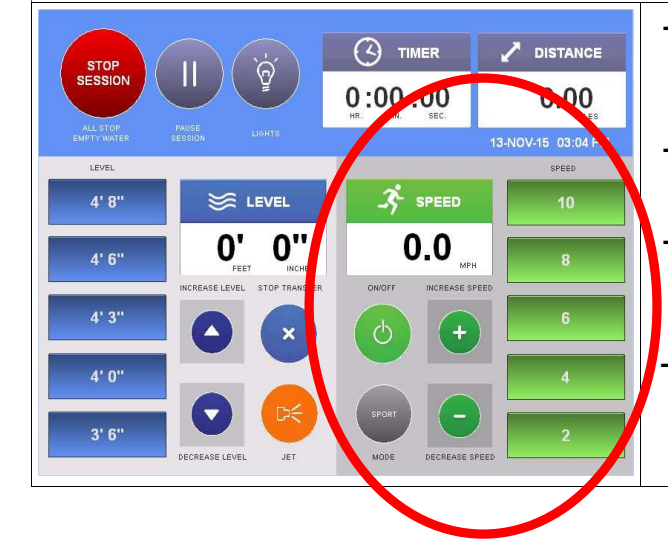
1. Warning Screen	
	**NOTE: Read and comply with all warnings and instructions listed on this screen before continuing.** **Press the Confirm button to continue.**

2. Water Depth Menu



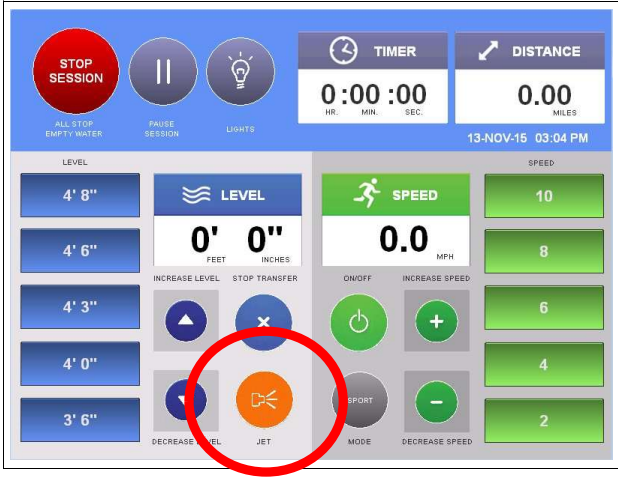
- **Pressing any of the pre-set water levels on the depth scale will fill the chamber to the selected depth.**
- **The + / – buttons fine-tune the water depth in 1” increments.**
- **The display window in the middle left of the screen indicates the actual water depth.**

3. Treadmill Menu



- **Press any of the pre-set speeds on the scale and then press On/Off to start the treadmill**
- **Use the + / – buttons to fine-tune the treadmill speed in .1 MPH increments.**
- **The display window in the middle right of the screen indicates the actual treadmill speed.**
- **Press the button marked On/Off to slow down and/or stop the treadmill at any time.**

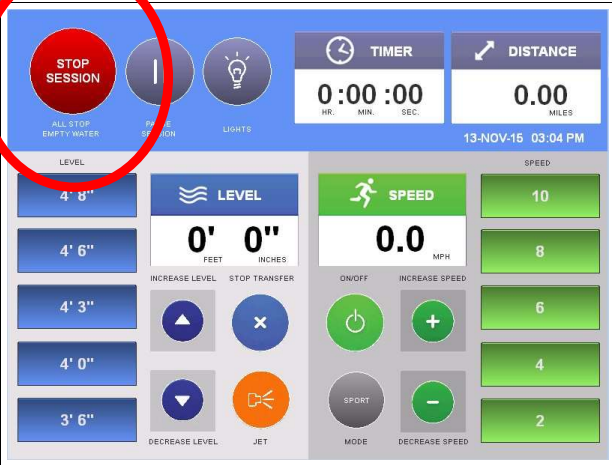
4. Resistance Jet



- **This menu shows the operation of the resistance jet.**
- **Press the Jet button to turn the resistance jet on and off.**

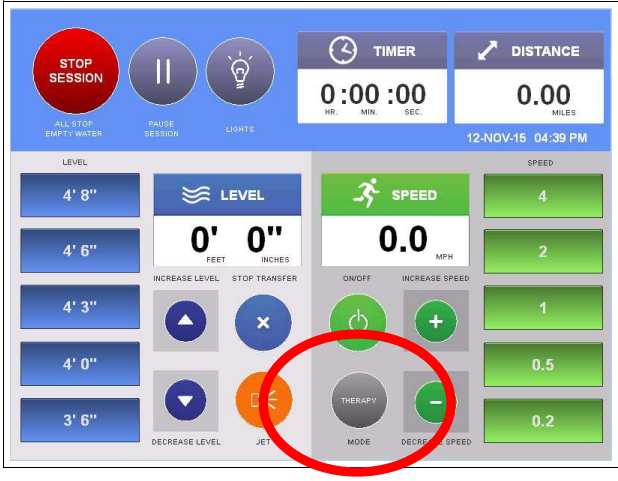
Note: The jet will not operate if the water level is below the jet outlet.

5. Stop Session



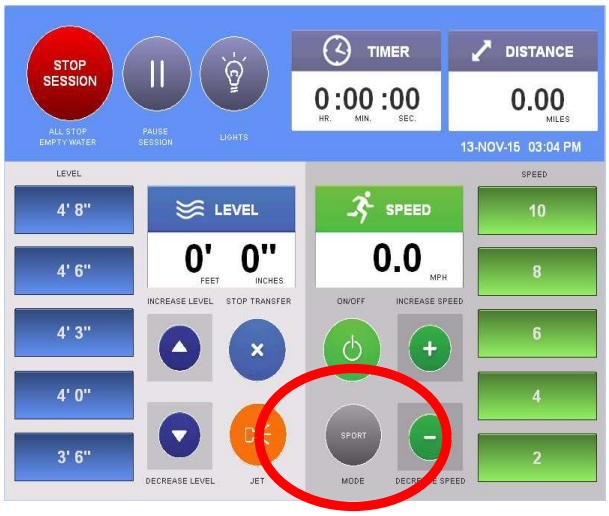
- **Pressing the stop session will stop operation of the treadmill and/or jet.**
- **The water will automatically drain from the 350 Treadmill.**

6. Therapy Mode



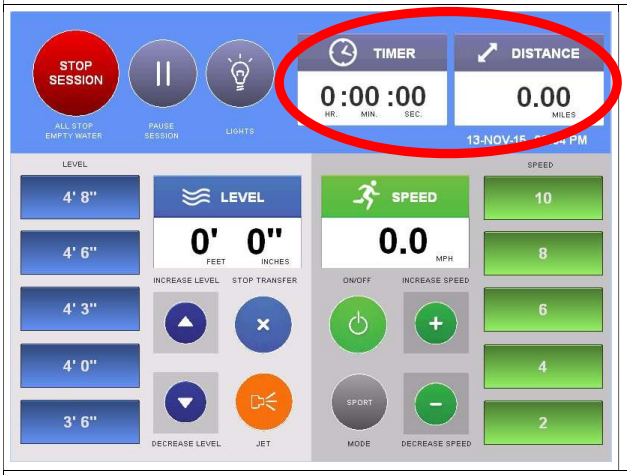
- **Press and hold the Mode button for 1 second to switch from Therapy Mode to Sport Mode.**
- **Therapy mode will display the quick speed settings as 0.2, 0.5, 1, 2, & 4.**

7. Sport Mode



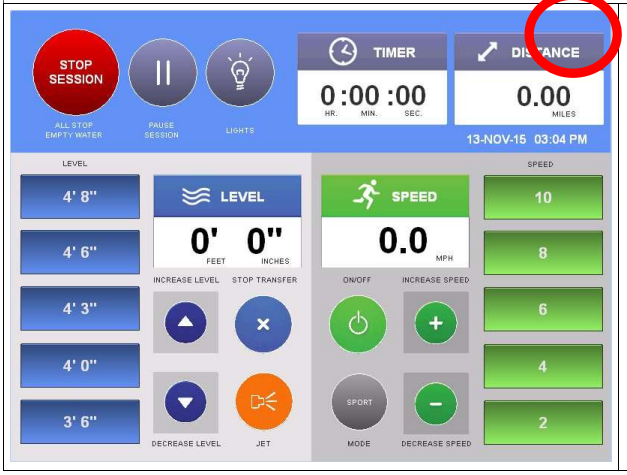
- **Press and hold the Mode button for 1 second to switch from Therapy Mode to Sport Mode.**
- **Sport Mode will display the quick speed settings as 2, 4, 6, 8, & 10.**

8. Timer and Distance

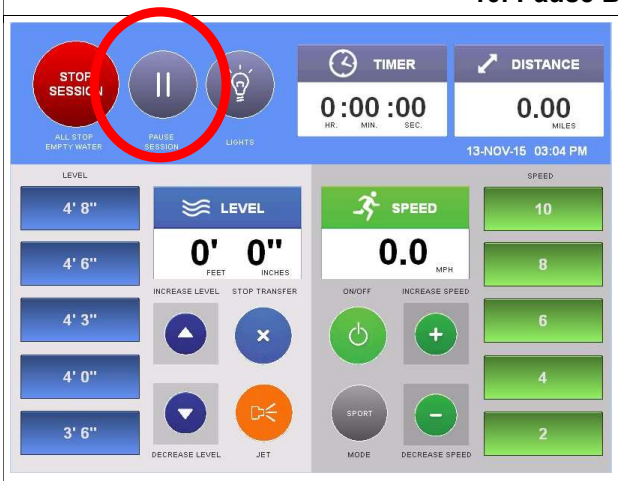
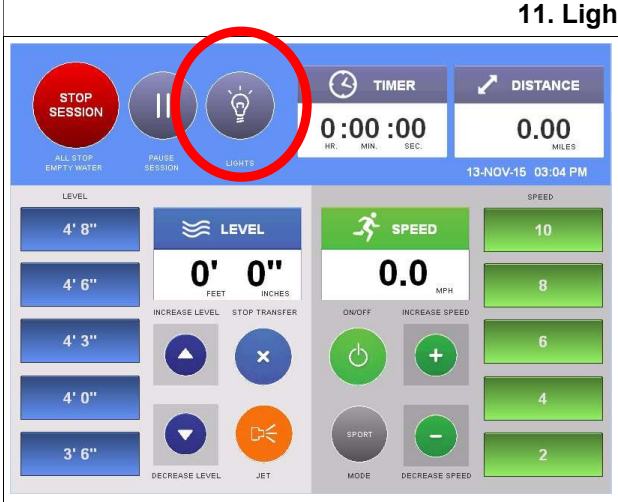


- **The Timer and Distance window displays the current run time and distance traveled on the treadmill.**

9. System Access



- **Pressing the top right corner of the main screen will access the other system menus**

10. Pause Button	
	- **Press the pause button to pause the treadmill, the jet, & the timer.**
11. Lights	
	- **Press the light button to turn on/off the blue LED light above the treadmill on the front glass panel**

Part 3

Overview of Operating and Maintenance Instructions

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3. Overview of Operation and Maintenance Instructions

3.1 HydroWorx 350 Series

Your HydroWorx 350 Series has been fabricated for you according to your particular specifications and will provide you years of reliable service if maintained properly.

Like any mechanical device, your HydroWorx System must regularly be inspected and have regular preventative maintenance procedures performed on the system.

The information contained within this manual will profile various components, systems and parts integrated into your HydroWorx 350 Series System. These profiles are for information purposes only and are not to be used for repair or replacement unless HydroWorx has specifically authorized the repair or replacement activity. Any unauthorized repair or replacement activity will void your limited warranty and may cause harm or injury to both the product and user.

In addition to the profile data, you will find diagrams, procedures, instructions, and other basic data for your reference.

3.2 Guidelines and Warnings

WARNING

Never use the HydroWorx System without supervision from a qualified aquatic therapy practitioner or exercise specialist.

WARNING

Never dive or jump into your HydroWorx System. Any unauthorized or extreme use may result in injury or death.

WARNING

Always have patients consult their physician or other medical professional before

WARNING

Unauthorized service, repair or replacement of parts will void your warranty and may damage your HydroWorx System or injure your patients or qualified practitioner.

WARNING

Never operate the treadmill with more than one person at one time on treadmill floor.

3.2 Guidelines and Warnings (Continued)

WARNING

Never operate the system without wearing the safety tether.

Electrical Components Installation & Maintenance Inspection Checklists and Warranty Activation Forms

Please reference the electrical installation checklists included in this manual for items to be completed by the project's qualified licensed electrician during installation and recommended annual inspections. *These checklists are critical to ensure installation meets the proper specifications of all referenced manufacturers and the owner should request copies of all completed forms from the electrical contractor before allowing the HydroWorx System(s) to be used. At a minimum, yearly inspections should be performed by a qualified licensed electrician to re-inspect all items noted on the checklists to ensure ongoing compliance.*

Documents included in the operational manual include:

3.3 ISTECH Warranty Activation Form

This is the form that comes with every ISTECH electrical panel that must be completed and emailed or mailed to ISTECH in order to activate warranty on the electrical panel and confirm the unit is installed per the original manufacturer's guidelines. This is the main power panel that controls functions on the HydroWorx System.

3.4 Plumbing and Electrical Schematics

These schematics provide general information for design considerations and installation requirements.

3.5 Resistance Jet

- **Adjusting Flow Rate**
- **Adjusting Flow Direction**
- **Pulsating Massage Hose**

3.5.1 Adjusting Flow Rate:

Grasp and turn the nozzle located in the center of the jet to decrease or increase the flow rate. Turning the nozzle clockwise will decrease the flow rate. Turning the nozzle counter-clockwise will increase the flow rate.

3.5.2 Adjusting Flow Direction:

Grasp and push the jet nozzle in any desired direction. The nozzle may be stiff and hard to move in the beginning of your use of the HydroWorx System, however it will become easier to adjust over time.

3.5.3 Pulsating Massage Hose:

1. Make sure jets are off before connecting.
2. Grab the end of the massage hose with the clip-on device.
3. Pull back on the clip and hold while pushing onto the jet nozzle.
4. When fully inserted onto the jet nozzle, push outside ring of the clip toward the jet to lock massage hose in place.

3.6 Treadmill System

3.6.1 Starting / Stopping and Adjusting Speed:

1. Select the desired mode: Therapy or Sport
2. Select desired preset speed
3. Press the on / off button to start the treadmill
4. Adjust speed up or down using the preset speed buttons in 0.1 increments



NOTE: Treadmill will always reset at 0 mph when turned off.

3.7 Filtration System

- **Changing the Filter Cartridge**
- **Checking Pump Strainer Baskets**

3.7.1 Changing the Filter Cartridge:

1. Turn off the power to the unit.
2. Using the isolation valves, isolate filter system plumbing from the system.
3. Open filter drain hose connection.
4. Open the air bleeder valve on top of the filter.
5. After draining the filter, remove the ring around outside of filter.
6. Lift top of filter off and remove cartridge.
7. Install new cartridge and put top of filter back on.
8. Reinstall ring around filter.
9. Close filter drain hose connection.
10. Return isolation valves back to normal operating position.
11. Air will begin to come out of air bleeder valve on top of filter. Once the air has stopped and water is coming out of the bleeder, close air bleeder valve
12. Turn on power to the unit.

3.7.2 Checking Pump Strainer Baskets

1. Turn off the power to the unit.
2. Using the isolation valves, isolate filter system plumbing from system.
3. Open the access lid on the pump, remove basket, and clean out any debris. Replace the basket back into the pump and inspect the O-ring found on the pump lid to be sure it is clean and fully intact. Screw the pump lid firmly back onto pump.
4. Return isolation valves to normal operating position.
5. Turn on the power to the unit.

3.8 Heater



WARNING!

- Do not operate without water
- Set the temperature to your desired setting using the digital control on front of the heater

3.8.1 Trouble Shooting:

Problem	Solution
No heat	Power to the heater is not turned on. Turn on power to the heater.
	Insufficient water flow through the heater – make sure the circulation pump is operating and the filter is clean. Filter pressure gauge should be in the 10-15 PSI range / if higher, the filter needs to be cleaned. Also check bypass valves to make sure enough water is flowing through the heater.
Element Lights are ON but the unit is not heating water	The heating element may need to be replaced.

3.9 Chemical Control

- **Water Chemistry**
- **Typical Supplies Needed to Maintain System**
- **Filling Brominator and Adjusting Feed Rate**

3.9.1 Water Chemistry

A Taylor Test Kit is supplied with the purchase of every unit. It is important to test the system water. Carefully follow the water test instructions to maintain the proper water chemistry balance. Chemical imbalances in your unit can cause the following:

- A ring on the 350 Treadmill around the water level
- Green water
- Excessive particles sticking to the system wall.
- In some cases, prolonged chemistry imbalance can cause damage to the heater, pumps, valves, and seals. This damage is not covered under the unit warranty.

It is also important to keep an accurate daily record of water chemistry. Profiles of your water chemistry components are on the following pages.



NOTE: WHEN FIRST FILLING YOUR SYSTEM, WE RECOMMEND BALANCING THE WATER CHEMISTRY IMMEDIATELY. TO DO THIS PROPERLY, FOLLOW THESE STEPS:

1. Check the pH level and set according to the test recommendations (7.4 – 7.6 is ideal).
2. Test Alkalinity and Calcium Hardness.
3. Set Alkalinity according to test recommendations (80 – 120 is ideal).
4. Set Calcium Hardness according to test recommendations (200 – 400 is ideal).
5. Shock System with Lithium Hypochlorite based shock (this chlorine based shock will kill living organisms and bacteria in the water).
6. Test water again later same day and again the next morning.

3.9.2 Typical Supplies Needed to Maintain System:

- Alkalinity Increaser
- Calcium Hardness Increaser & Decreaser
- Chlorine based Shock with Calcium Hypochlorite in it.
- pH Decreaser
- Bromine Tablets

3.9.2 Typical Supplies Needed to Maintain System (Continued):

- Metal Free
- Spa Stain & Scale
- System Perfect
- Garden hose & submersible pump
- (1) Spare system filter (call HydroWorx to order 1-800-753-9633)
- Gentle cleaner and non-abrasive sponges
- Test strip kit to check water balance
- Stainless steel cleaner, glass cleaner, and non-abrasive cloths

3.9.3 Filling Brominator:



CAUTION: EYE PROTECTION AND GLOVES ARE REQUIRED. AVOID BREATING FUMES!



1. Carefully remove cover cap.
2. Slowly add 1” Bromine tablets to fill to 1/3 capacity. Try to limit water splashing as much as possible.
3. Replace cover cap securely.

3.9.4 Adjusting Brominator Feed Rate:

1. Upon initial system fill, set variable setting valve on outside of brominator to setting 1.5.
2. After testing water with Taylor Test Kit, adjust Bromine feed rate by increasing or decreasing the variable setting valve.

3.10 Electrical Panel (E-Stop) System

▪ 3.10.1 How to Reset E-Stop



NOTE: **An E-Stop button is located on the front door of the electrical panel connected to the Plant Pack. This red button has a removable key that will lock out the controls of the unit, but will keep the water circulating. To reset, turn the key until the red button releases.**

3.11 Draining and Filling the Storage Tank

3.11.1 Draining:

1. Turn off the power to the unit
2. Open drain valves routed to sump pit or floor drain(s) (if applicable)
3. Lower sump pump into storage tank and route ¾" hose to sump pit or floor drain
4. Drain and vacuum 350 Treadmill and Plant Pack



NOTE: Be sure to monitor sump pit or floor drains when draining the system. If these do not keep up with the draining water flow, partially close the system drain valves to slow down the water flow.

3.11.2 Filling:

1. Close all drain valves.
2. Open valve in water line or from facility water source to storage tank.
3. Bleed out any trapped air inside filter housing (see Filtration System section – Changing the Filter Cartridge).
4. Turn on the power to the unit.

3.12 System Maintenance Records

- **Stainless Steel Care**
- **System Maintenance Schedule**
- **Daily Chemical Logs**
- **Weekly Chemical Logs**
- **Weekly Maintenance Records**
- **Monthly Maintenance Records**
- **Four (4) Month Maintenance Records**

All logs and records can be found on the HydroWorx provided removable USB drive.

3.12.1 Stainless Steel Care

Stainless steel cleaning & Maintenance

Stainless steel is a durable, corrosion - resistant chromium/nickel alloy steel with excellent luster. However, it is not rustproof - particularly within the harsh environment of a system. Two issues are at play here.

1. Chlorine and bromine sanitizers are highly caustic chemicals for stainless steel.

2. Heat and humidity enhance the corrosiveness of these chemicals.

Regular cleaning is the best way to prevent corrosion and add to the service life for stainless steel handrails, support bars, exterior decking, and any other system equipment. The goal of the cleaning & maintenance program should be to keep the protective chromium oxide layer of the stainless steel intact.

Exposed stainless steel in your HydroWorx System includes the following:

- **Interior handrails**
- **All outside trim**
- **HMI Panel**

1. Clean these components frequently (at least one time per week) with a stainless steel cleaner and water.

2. Inspect equipment frequently and if you notice discoloration, tarnish, or water stains, increase the frequency of your fresh water rinses to reduce accumulated chemicals.



CAUTION!

Risk of material damage

- **Do not use steel wool or sandpaper, mineral acids, bleaches, or chlorine cleansers.**

- **Do not add chlorine to the system in close proximity to stainless steel equipment. Sanitizers should be added as far away as possible from stainless steel.**

3.12.2 System Maintenance Schedule

Daily:

- ✓ **Check Bromine and pH water chemical levels.**

Weekly Maintenance:

- ✓ **Check Bromine tablet level in brominator.**
- ✓ **Shock water with Lithium Hypochlorite Shock.**
- ✓ **Check Alkalinity and Calcium Hardness water chemical levels.**

Monthly:

- ✓ **Test the main GFCI breaker.**
- ✓ **Clean or change the circulation filter.**

Every Four (4) Months (or as usage dictates):

- ✓ **Drain and clean the unit.**

Part 4

Maintenance Instructions

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4.1 Operation and Maintenance

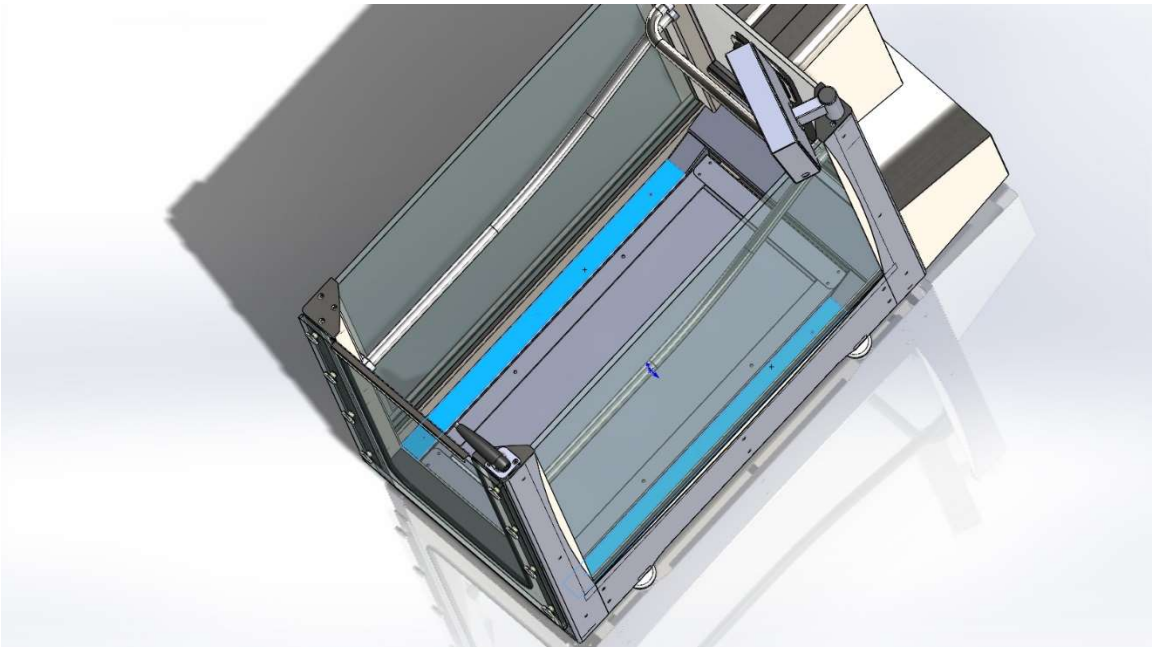
4.1.1 Operation of the System When Not in Use

When the system is left overnight or over the weekend, it is advisable to leave the power switched on and the recirculation system running. This ensures that the water will continue to be filtered, heated, and ready for the next day. If the power is off or if the tank is drained, one of the following procedures must be used when restarting the system.

4.1.2 Enable the Treadmill Chamber to be Cleaned:

1. Open treadmill chamber door.

2. Remove two (2) bolts from each of the treadmill side rails and remove rails (highlighted below in blue).

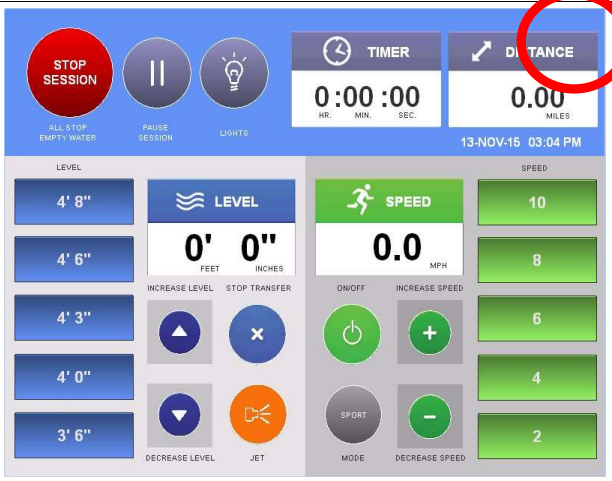
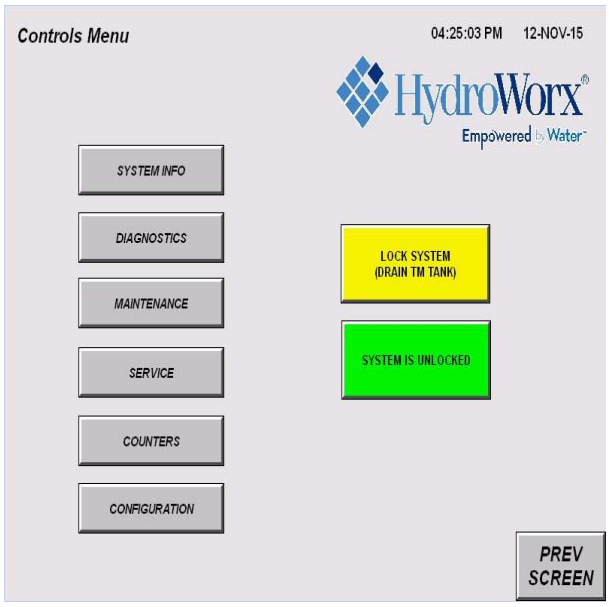


3. Rotate clockwise 90 degrees, using the key as a handle. Pivot the treadmill to a full upright position and it will lock with a safety bail.

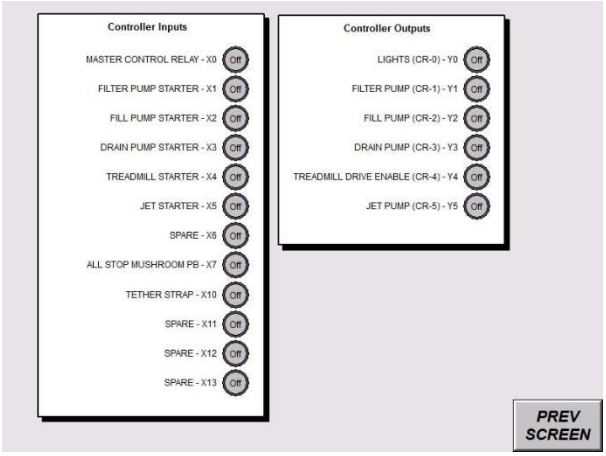
4. Clean and remove debris from chamber. The use of a wet/dry vacuum cleaner is recommended.

WARNING! **Do not run treadmill unless locked with side rails secured. Failure to observe this could result in injury to user, or damage to the equipment.**

4.1.3 Touch Screen Maintenance Mode

1. Settings & Administration	
	- The settings and administration screen is accessed by touching the main menu in the upper right hand corner. - The screen will enter sleep mode after 60 minutes of non-use. Touch the screen anywhere to awaken it.
	- The date and time can be changed by accessing the Date and Time Setup tab. - The Annunciation tab allows the user to turn off/on the treadmill warning lights that blink to inform the user the treadmill is starting. - The Safety Limits “Caution” tab allows the user to change the preset depths. - The Level Controls tab allows the user to reset and calibrate the water depth sensor.

1. Settings & Administration



- This screen displays the system condition and informs the user of the current system status.

Part 5

Installation Instructions

HYDROWORX

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Access & Serviceability

Recommended minimum architectural, structural, mechanical, electrical, and plumbing design criteria identified in this design and installation guide must be maintained for future serviceability of the HydroWorx 350. **Any design that does not meet the minimum requirements must be approved by HydroWorx.**

Site Preparation

The owner/owner's representative/owner's contractor is responsible for preparation of the site prior to delivery. The system's final location including all utilities (incoming power, water, sanitary, etc.) required for system operation shall be designed and installed by the owner/owner's representative/owner's contractor. For more details about system requirements, please refer to the subsequent technical specifications.

System Location

The water treatment area of this HydroWorx system is designed to accommodate one (1) patient/athlete during treatment sessions in the 350 Treadmill. The recommended ***minimum*** dimensions for the 350 Treadmill location are 7'5" wide by 14'8" long by 10' high allowing adequate installation and deck space around 350 Treadmill. The recommended ***minimum*** dimensions for the plant pack and storage tanks are 5'8" wide by 14'8"' long by 10' high. Additional space may be required for: cabinetry, storage, showers, changing rooms, etc.

Floor drain(s) are recommended to be incorporated into the design of the system area. The finished floor MUST be flat/level, smooth, consistent at all edges, and the floor should slope away from walls towards the floor drain(s).

Adequate airflow and temperature control is required primarily to maintain a comfortable temperature for the occupants, and to protect against adverse conditions. Water in the Storage Tank is generally kept between 92-94 degrees Fahrenheit.

The hydrotherapy room should be designed to maintain an ambient temperature typical for occupancy (79º - 85º F) and for maintenance of the pool equipment. The information below may be utilized for heat discharge in determining the appropriate HVAC design for final location of the 350 series equipment. The HydroWorx 350 Series fully equipped, includes the following:

1 – 5.5 kW Electric Heater
1 – 3.5 HP Front Therapy Jet Pump 7,350 BTU/hr)
1 – 5 HP Treadmill Motor (12,735 BTU/hr)
2 – 3.5 HP Transfer Pumps (7,350 BTU/hr X 2)
1 – 3/4 HP Maintenance Pump (1,910 BTU/hr)

Equipment Size and Weights

350 Treadmill – 35” W X 90” L X 73” H // 1,650 LBS Dry // 6,863 LBS Wet
Plant Pack – 34” W X 72” L X 76” H (89” of room height needed) // 765 LBS Dry // 974 LBS Wet
Storage Tanks – (2) 500 Gallons, 1000 Gallons Total // 31” W X 68” L X 64” H // 400 LBS Dry // 8,740 LBS Wet

Utilities

Water Supply: The owner shall provide a water supply line or ¾” diameter hose bib to fill the Water Storage Tank.

Electrical Service: The owner shall provide an electrical service designed for the HydroWorx control panel requirement which is:

Standard: 208-240 Volt, 60 Hz , 100 amp, 1-phase, 4-wire configuration (L1, L2, Ground, Neutral)

Sanitary: The owner shall provide a sump pit with pump or floor drain(s) to accommodate 1,000 gallons via gravity through a 2” PVC pipe or ¾” diameter hose. The HydroWorx maintenance pump CANNOT be used to drain the pool or it will void the components warranty.

Owner/General Contractor Scope-of-Work:

Construct the system space per the owner's design documents (if any) and in accordance with this HydroWorx Design & Installation Guide.

The 350 Treadmill is a 35" wide by 90" long by 73" high. The Plant Pack is 34" wide by 72" long by 76" high. A minimum 36" wide by 84" high path from the anticipated off-loading point into the system space must be in place at time of delivery. The unit must remain upright at all times during delivery to avoid damaging the integrated treadmill and the 350 Treadmill. **The unit cannot be turned on its side.**

Customer/Customer's Representative/Customer's General Contractor is responsible for receiving and unloading the unit components and equipment based on the delivery date noted on the **Delivery Confirmation Form (DCF).**

The unit will be delivered to the project site and it is Customer/Customer's Representative/Customer's General Contractor's responsibility to unload and set the 350 Treadmill, Plant Pack, & Water Storage Tank into its final location. The system will be protected and crated for shipping and delivery. Below you will see a picture of the typical loading method where a lull/forklift is used to remove the unit.



Following placement, level the 350 Treadmill and Storage Tank using the adjustable feet.

Upon completion of the electrical and plumbing requirements, clean the 350 Treadmill using glass cleaner and a non-abrasive cloth. Fill the Water Storage Tank and complete the site analysis checklist procedures necessary to place the

system into service. **This form must be completed and returned to HydroWorx before the customer in-service training can be scheduled.**

Typical Electrical Contractor Scope-of-Work:

Provide all labor, materials, and equipment to connect the required 208-240 Volt, 60Hz, 100 Amp, 1-phase, 4-wire configuration (L1, L2, Ground, Neutral) to the HydroWorx control panel following the provided electrical installation diagram placed in the electrical control panel.

Install the provided interconnecting cables from the HydroWorx control panel and terminate the feed to each of the following system components after checking the voltage of the 100 amp service feed:

1 - Low voltage control cable from HydroWorx control panel to the 350 Treadmill low voltage junction box.
2 - Treadmill power cable from the HydroWorx control panel to the treadmill motor junction box.
3 - #6 ground wire from the HydroWorx panel to the marked ground lug on the 350 Treadmill.
4 - HMI touch screen and cable through routing holes and into low voltage junction box.

Please refer to the subsequent technical specifications.

Gauge of the ground wire of any appliance must be as large as the hot wire(s) so it is capable of efficiently conducting electricity away from the device in the event of a short circuit or stray current, as well as prevent GFCI nuisance tripping.

Electrical Components Installation & Maintenance Inspection Checklists and Warranty Activation Forms

Please reference the electrical installation checklists included in this installation guide for items to be completed by the project's qualified licensed electrician upon installation and recommended follow-up yearly inspections. ***These checklists are critical to ensure installation meets the proper specifications of all referenced manufacturers and the owner should request copies of all completed forms from the electrical contractor before allowing the HydroWorx systems(s) to be used. At a minimum, yearly inspections should be performed by a qualified licensed electrician to re-inspect all items noted on the checklists to ensure ongoing compliance.***

The documents included in this installation guide are the following:

ISTECH Warranty Activation Form

This is the form that comes with every ISTECH electrical panel that must be completed and faxed or mailed to ISTECH to activate warranty on the electrical panel and confirm the unit is installed per the original manufacturer's guidelines. This is the main power panel that controls functions on the HydroWorx system on the HydroWorx 350 system

Typical Plumbing Contractor Scope-of-Work:

Provide all labor, materials, and equipment necessary to install interconnecting plumbing between each HydroWorx 350 component as required by the owner's design documents (if any) and in accordance with this HydroWorx Design & Installation Guide.

Provide means to drain the storage tank using a rated sump pump with a ¾" diameter hose or PVC piping via gravity to a sump pit or floor drain(s) to accommodate 1,000 gallons of water.

All piping, fittings, valves, etc. shall be provided and installed by the plumbing contractor. Unless otherwise noted, all conveyance piping, fittings, valves, etc. shall be schedule 40 PVC (pressure type) or schedule 40 flex pipe fastened with primer and heavy-bodied pipe glue. Threaded fittings shall be connected using silicon sealant (do not use Teflon tape or plumbing pip-thread sealer).

After the electrician and plumber have completed all connections, fill the pool to hydrostatically test plumbing installed. If the pool is left up and running, add shock and other chemicals to balance and maintain the water chemistry.

Minimum Start-Up & Testing Requirements:

- Set up the HydroWorx controls system
- Complete the HydroWorx Site Analysis Checklist and return it to the HydroWorx project management department for review at projectmanagement@hydroworx.com prior to starting or testing the system. **Customer in-service training cannot be scheduled until this is completed**
- Prior to filling the pool with water, confirm connectivity of all pool components
- Check voltage of incoming electrical supply

- Complete ALL Electrical Components Installation & Maintenance Inspection Checklists and Warranty Activation Forms
- Complete ALL plumbing per the Installation Guide
- Leak test all plumbing connections and confirm valve configuration
- Clean the 350 Treadmill stainless steel with stainless steel cleaner and a non-abrasive cloth
- Clean the 350 Treadmill with glass cleaner and a non-abrasive cloth
- Fill Water Storage Tank

Following completion of the start-up and testing work identified on the HydroWorx Site Analysis Checklist and verified by HydroWorx, a HydroWorx representative will schedule a site visit with the owner to perform training and place the pool into service.

HydroWorx Design & Installation Checklist

This document is designed to provide steps to improve the coordination of the HydroWorx pool(s) in the design and construction phase of the project.

Step 1

- ☐ **Have you had a conference call with a HydroWorx design & construction representative to discuss design requirements, scope of work, lessons learned etc.?**

  This is critical to have as EARLY as possible in the project. It is very useful in understanding the connection between Architectural, Structural, Mechanical, Electrical, Plumbing and the design & installation requirements of a HydroWorx System.

Step 2

- ☐ **Have you provided HydroWorx the pool room design for review and comment?**

Step 3

- ☐ **Have you provided HydroWorx with project approved final drawings (if any)?**

Top Lessons Learned

1. **Site and location requirements are not adhered to creating access issues for delivery, installation, future service, or component failure due to incorrect design.**

2. **Scope of work is not clearly defined resulting in change orders that impact the owner.**

3. **"Verify / Coordinate with manufacturer" is a common note on drawings that lead to errors and omissions or change orders that can be avoided by verifying the design requirements. HydroWorx believes it critical that we be involved early in the design / development phase given the structural and foundation impacts of our product.**

4. **Installation of the HydroWorx system into an existing building – It is important to identify any criteria that are applicable to existing buildings. If not addressed EARLY in the design process they are often not documented on building "As-Builts". The potential problem exists of running into unexpected issues that affect the system location planning and clearances and require last minute changes. It is critical that your engineering drawings (if any) identify that all piping, electrical service, structural, mechanical etc. is clear or re-routed in the area proposed for the system location.**

5. **Copper or salt water treatment systems - These systems will have a negative reaction with the stainless steel components which will result in voiding the warranty for the affected components. Please consult with HydroWorx on any desired alteration to the standard bromine water quality system.**